GRC103 — Advanced Cybersecurity Risk Quantification and Control Investment Analysis

Week 2 Practical Lab Report

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1. Overview

This assessment looked at three security risks affecting SecureBank Financial's SecureMobile banking application: API authentication bypass, database injection, and session hijacking.

The aim was to estimate the financial exposure associated with these risks, evaluate potential security controls, and identify where the organisation should focus its investment.

Based on the calculations, the combined baseline Annualized Loss Expectancy (ALE) is $35 million. Database injection accounts for most of this exposure, making it the main priority for further investigation and treatment.

One important point to note is that the lab brief provides exploit probabilities but does not specify their time period. The assessment treats them as annual probability proxies, so the figures are estimates rather than confirmed annual losses.

2. Key findings

Vulnerability	Estimated ALE	Priority
Database Injection	$31,250,000	High
Session Hijacking	$3,000,000	High
API Authentication Bypass	$750,000	Medium
Total	$35,000,000	

Database injection presents the largest estimated financial exposure. The assessment estimates that 500,000 customer records could be at risk, with a cost of $250 per record.

Session hijacking is also a significant concern because it could affect 5,000 simultaneous sessions, with an estimated loss of $1,500 per account.

API authentication bypass has a lower estimated ALE than the other two risks, but the potential loss from a single incident is still substantial.

3. Proposed security controls

The assessment considered three controls and their associated costs.

Proposed control	Initial investment	Annual maintenance	Stated effectiveness
Advanced API Security Gateway	$350,000	$50,000	90%
Web Application Firewall (WAF)	$150,000	$25,000	75%
MFA Enhancement	$200,000	$30,000	95%
Total	$700,000	$105,000	

The combined initial investment is $700,000, with annual maintenance costs of $105,000.

The effectiveness percentages are those provided in the lab brief. However, the WAF's specific target vulnerability still needs confirmation before its financial benefit can be calculated reliably.

4. Financial evaluation

The current calculations show the following results for the two controls with confirmed risk mappings in the workbook.

Measure	API Security Gateway	MFA Enhancement
Baseline ALE for the assessed risk	$750,000	$3,000,000
Estimated annual risk reduction	$675,000	$2,850,000
Residual ALE	$75,000	$150,000
Estimated payback period	6.7 months	26 days
Five-year ROI	462.5%	3,971.4%

Supporting Excel workbook: SecureBank_Risk_Quantifictaion Open Excel workbook

These results suggest that both controls could offer financial benefits under the assumptions used in the model. MFA has the higher calculated return, while the API Gateway directly addresses the API authentication bypass risk.

These are modelled reductions in expected loss, not guaranteed cash savings. The payback and ROI figures depend on the assumed effectiveness, stable annual benefits, the five-year evaluation period, and the treatment of maintenance costs. They should be validated before being used to approve expenditure.

5. Recommendations

Based on the assessment, I recommend the following actions:

Prioritise database injection. It accounts for the largest share of the estimated risk exposure and warrants further technical investigation and appropriate database security measures.

Evaluate MFA enhancement. The model shows a strong potential financial return for reducing session hijacking risk.

Proceed with the API Gateway evaluation. The estimated risk reduction and payback support further consideration of the proposed control.

Confirm the WAF's scope. Identify which assessed vulnerability or vulnerabilities it will address before including its risk reduction in the financial comparison.

Validate the assumptions. Confirm the probability time period, control effectiveness, and financial impact estimates before making final investment decisions.

Track implementation. Use the proposed 30-day, 90-day and one-year timeline to plan implementation and review progress.

6. Conclusion

The assessment highlights database injection as SecureBank Financial's largest modelled risk, followed by session hijacking and API authentication bypass.

The proposed controls require a combined initial investment of $700,000. The current financial model supports further evaluation of MFA and the API Gateway, while the WAF's benefits remain unconfirmed until its target risk is established.

The next step is to validate the assumptions, confirm the WAF mapping, and use the results to support a risk-based decision on security investment.